

Fusing Computer Vision with Avian Morphology to Predict Flight Dynamics

The AI Process

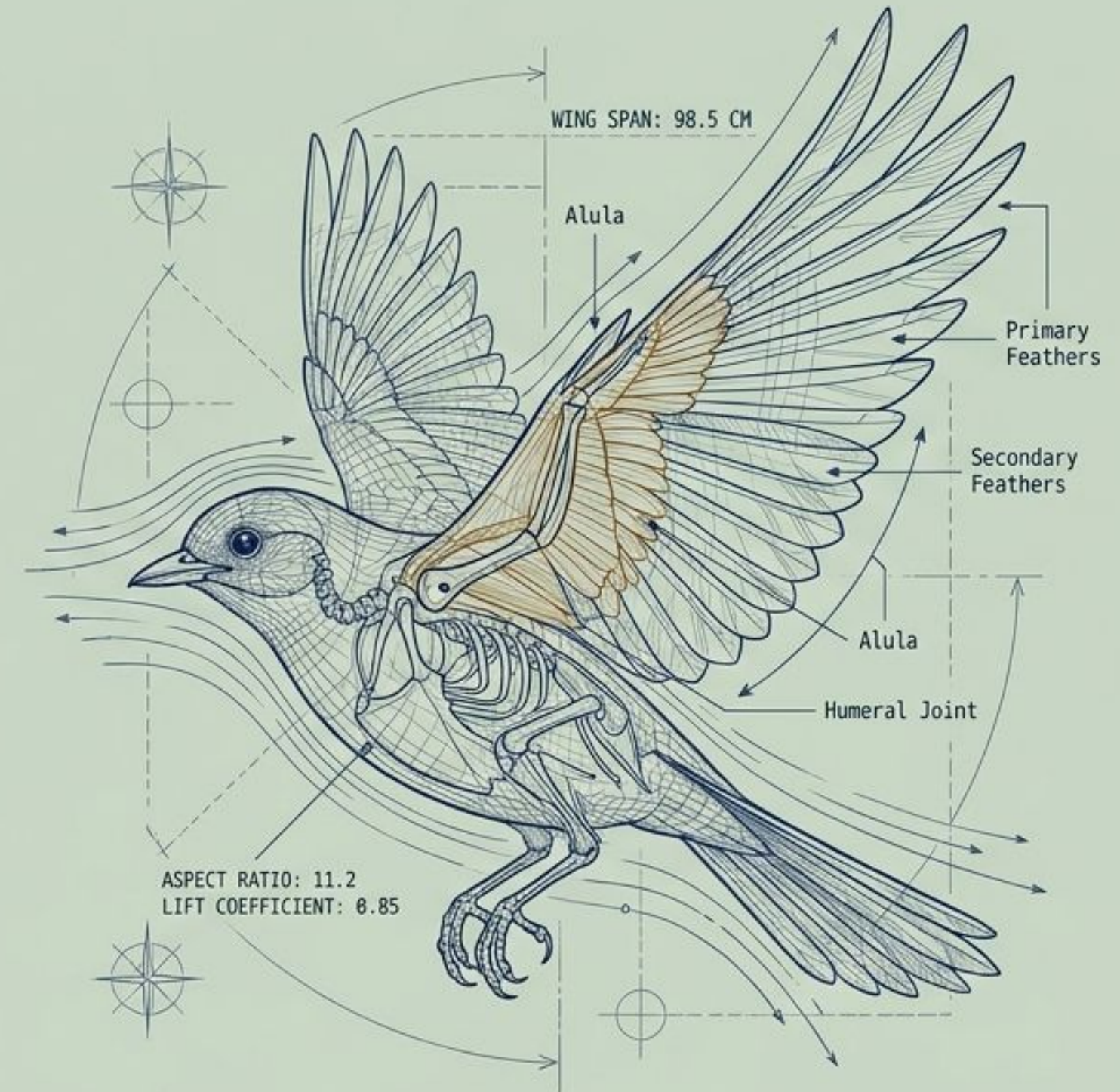
Motivation: Bird body structure and flight behavior are inextricably linked. Machine learning provides a powerful, quantifiable way to explore and explain this biological relationship.

Main Idea: A multimodal AI pipeline that ingests a user-uploaded bird image, predicts its taxonomic family, links it to tabular physical measurements, and computationally generates its likely flight style and wing model.

Results Briefly:

- Successfully adapted MobileNetV2 to classify five target bird families.
- Engineered morphological features (wing loading, aspect ratio) predicting speed with an R^2 of 0.956 (Ridge Regression).
- Achieved 99.6% accuracy classifying unsupervised K-means clusters into biologically accurate flight styles.

Biological Interpretation



Pawel Obrycki

Literature review

This project uses transfer learning with MobileNetV2¹ for image-based bird-family classification, morphology-based feature engineering from AVONET² data, regression to estimate a speed index, and K-means clustering³ to discover flight-style groups. The literature supports this combination because CNN transfer learning is effective for small image datasets, and avian morphology research⁴ shows that wing loading, aspect ratio, and body mass are closely related to flight behavior.

1 - MobileNetV2: Inverted Residuals and Linear Bottlenecks - Mark Sandler - [link](#)

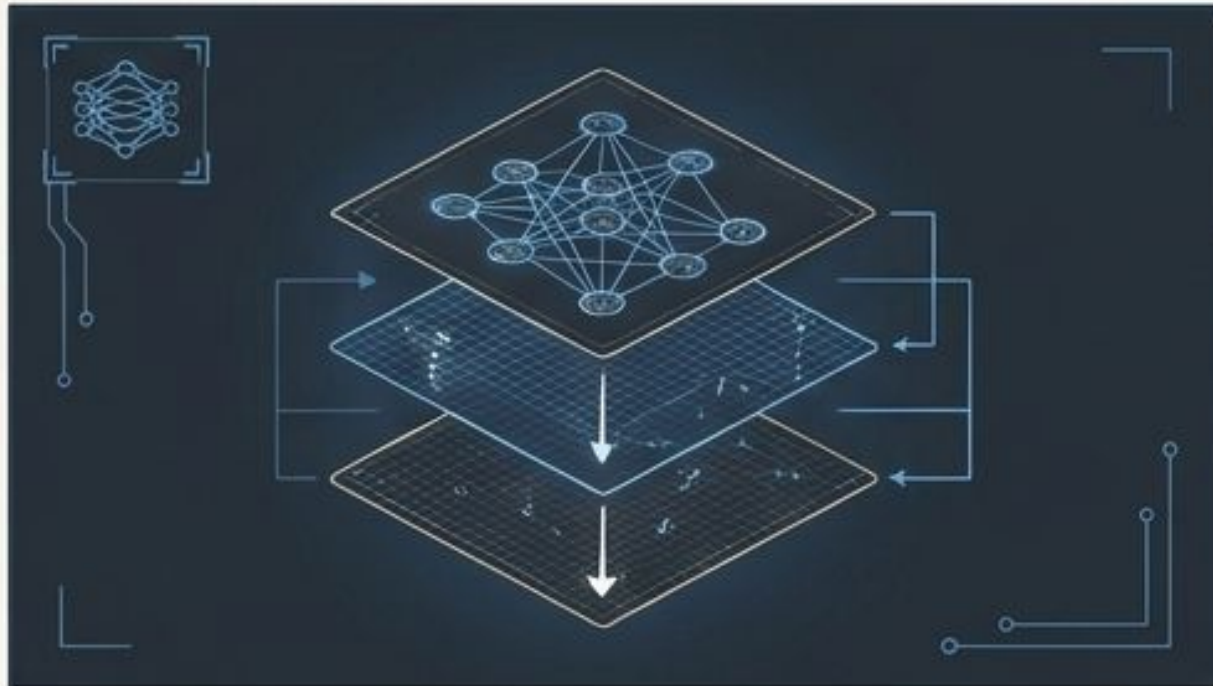
2 - AVONET: morphological, ecological and geographical data for all birds - Joseph A. Tobias - [link](#)

3 - Some methods for classification and analysis of multivariate observations - J. MacQueen - [link](#)

4 - Wing morphology, flight type and migration distance predict accumulated fuel load in birds - Orsolya Vincze - [link](#)

The Methodological Landscape: Core AI Concepts Applied

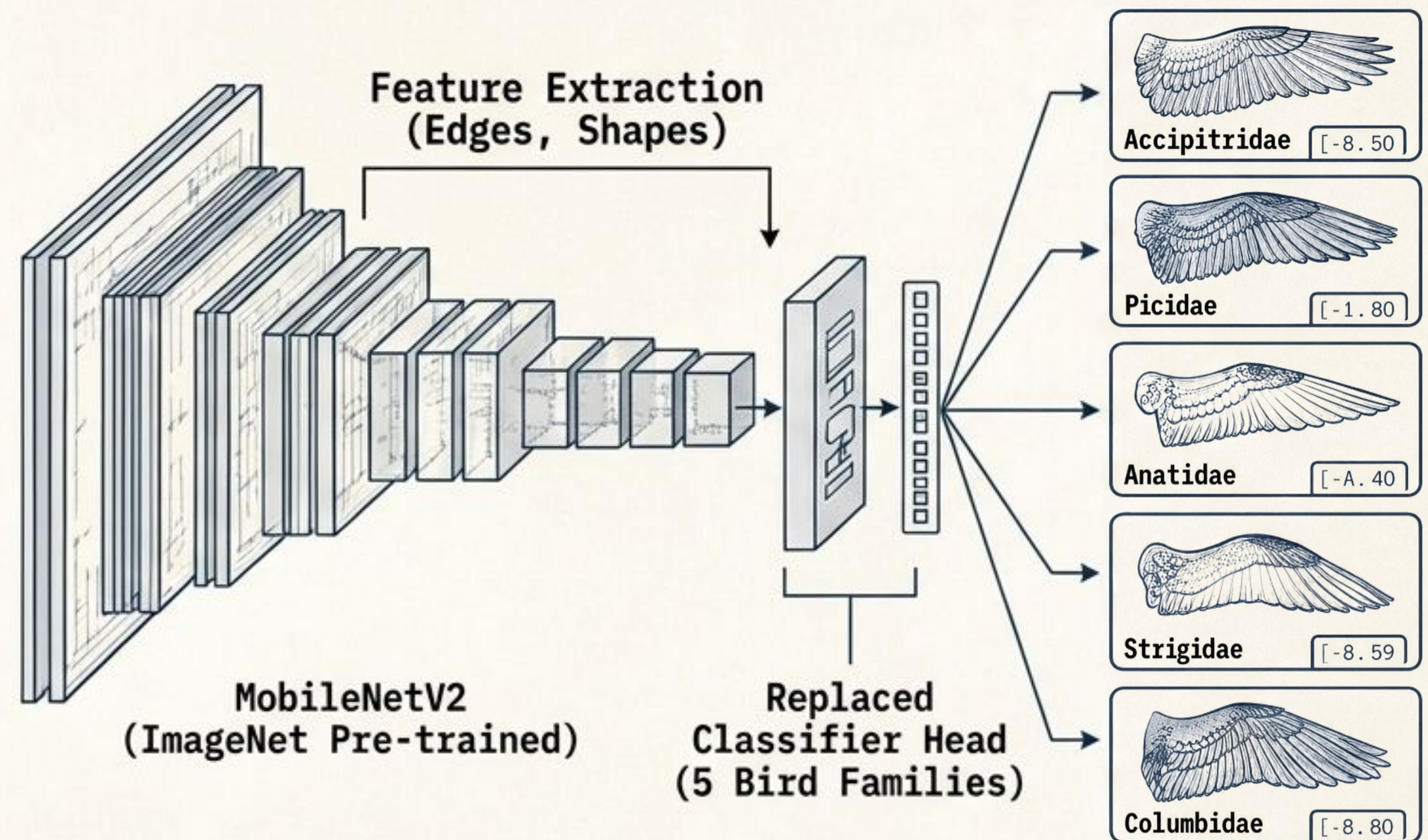
- Transfer Learning



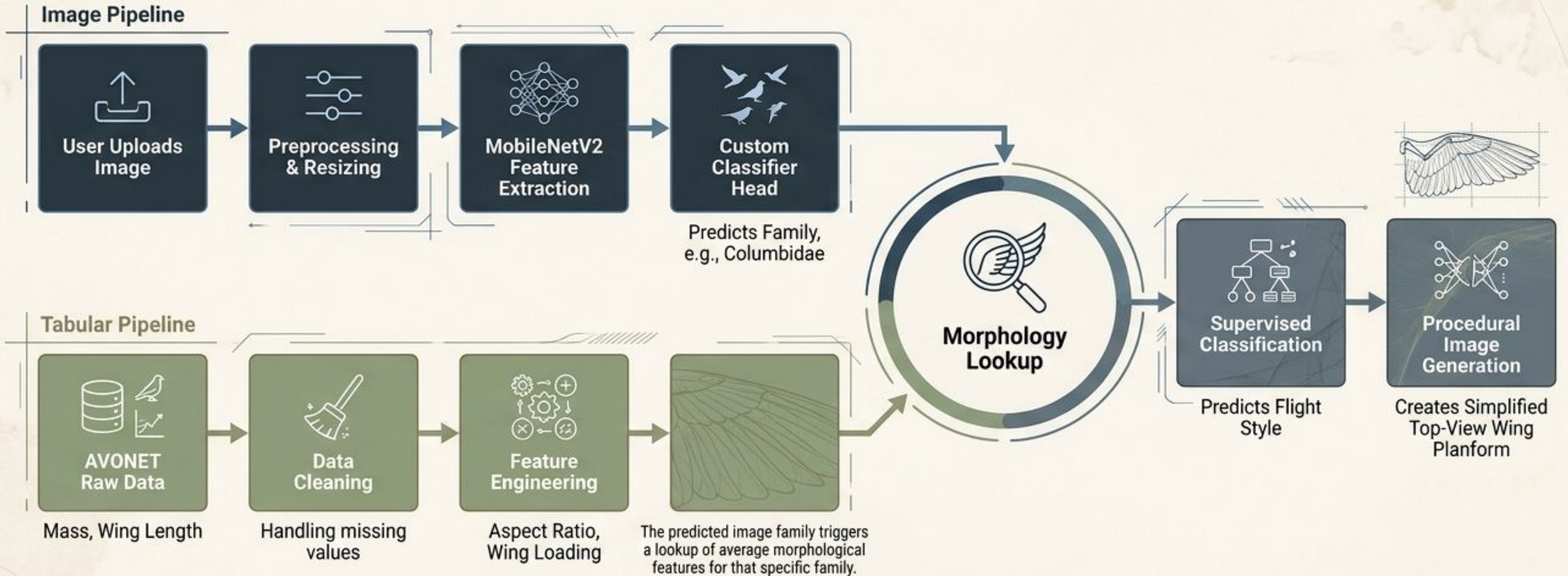
Method: Using models pre-trained on vast datasets for specific, smaller tasks.

Application: Utilizing MobileNetV2 (trained on ImageNet) to extract general visual patterns (edges, shapes) and replacing the classifier head to detect five specific bird families.

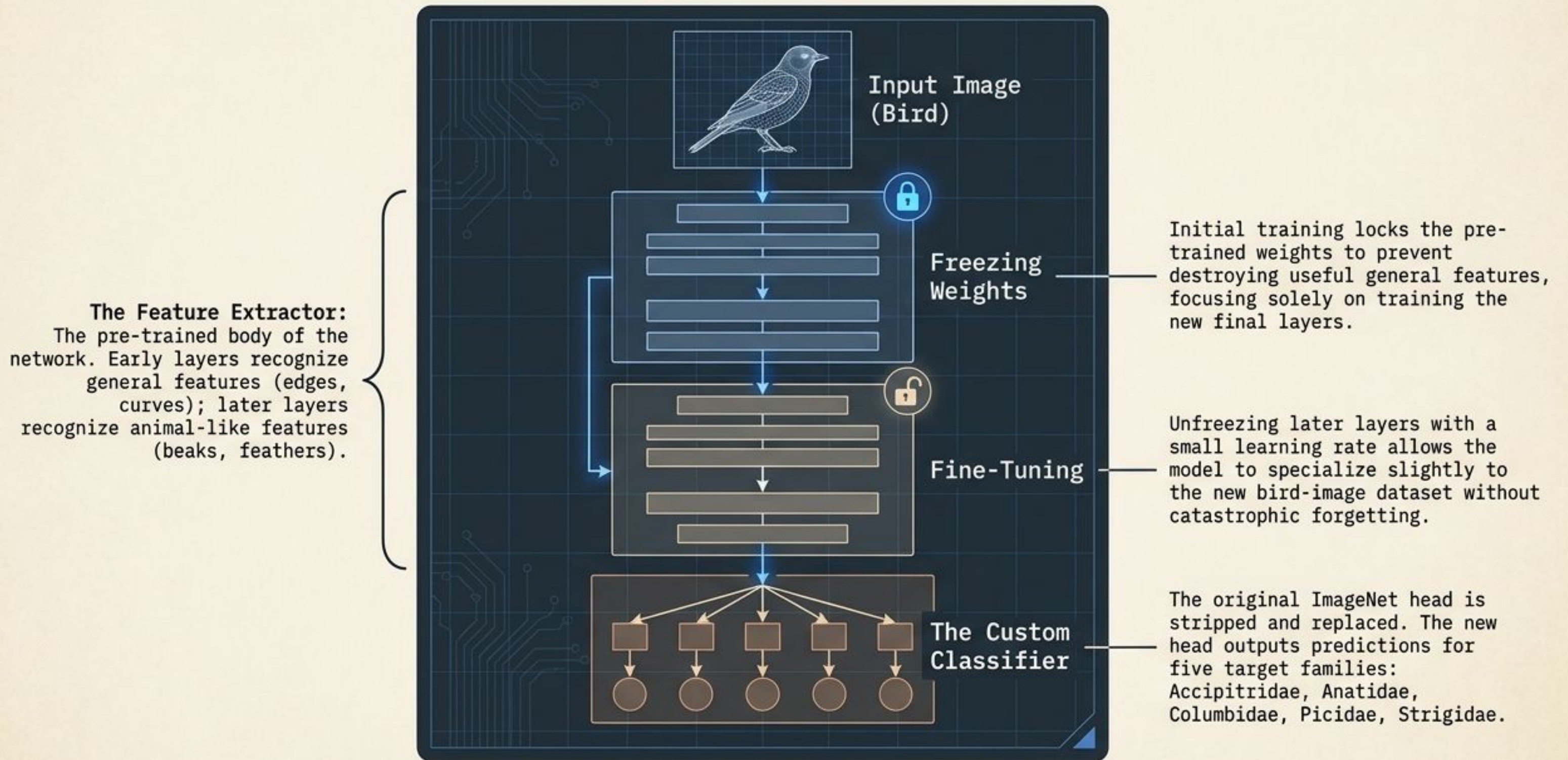
Transfer Learning



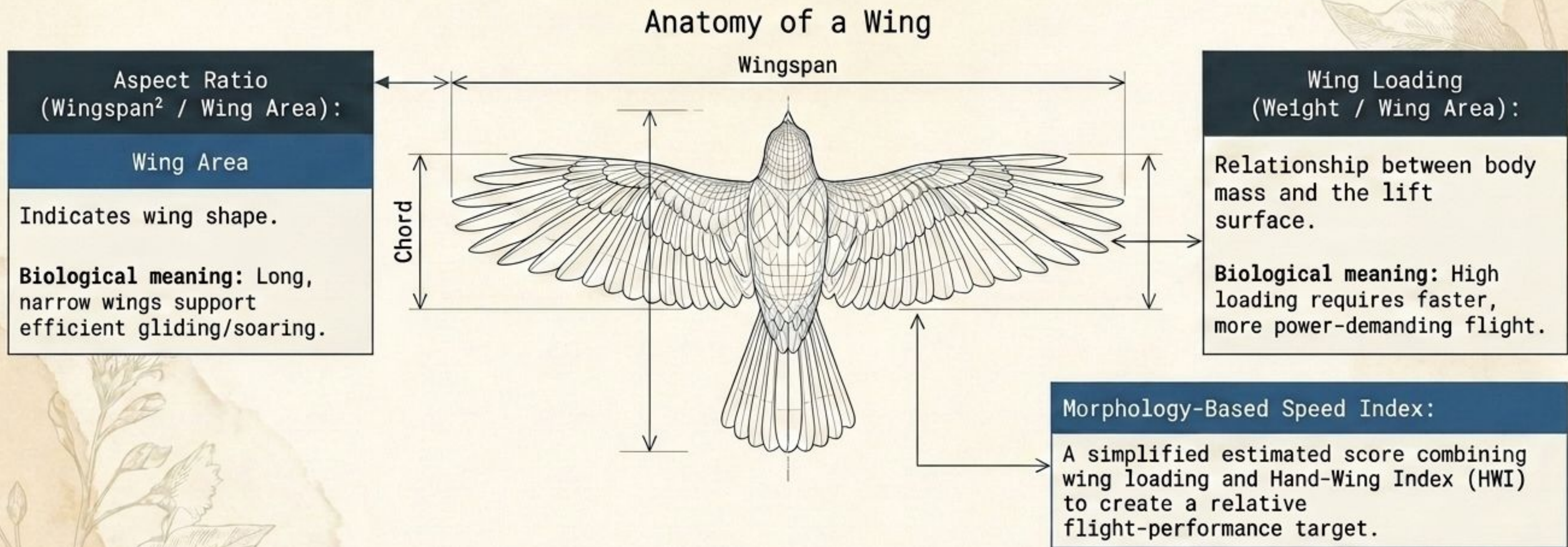
The Multimodal Architecture: Bridging Pixels and Physicality



Adapting MobileNetV2 for Avian Recognition via Transfer Learning



Feature Engineering: Translating Raw Measurements into Flight Dynamics



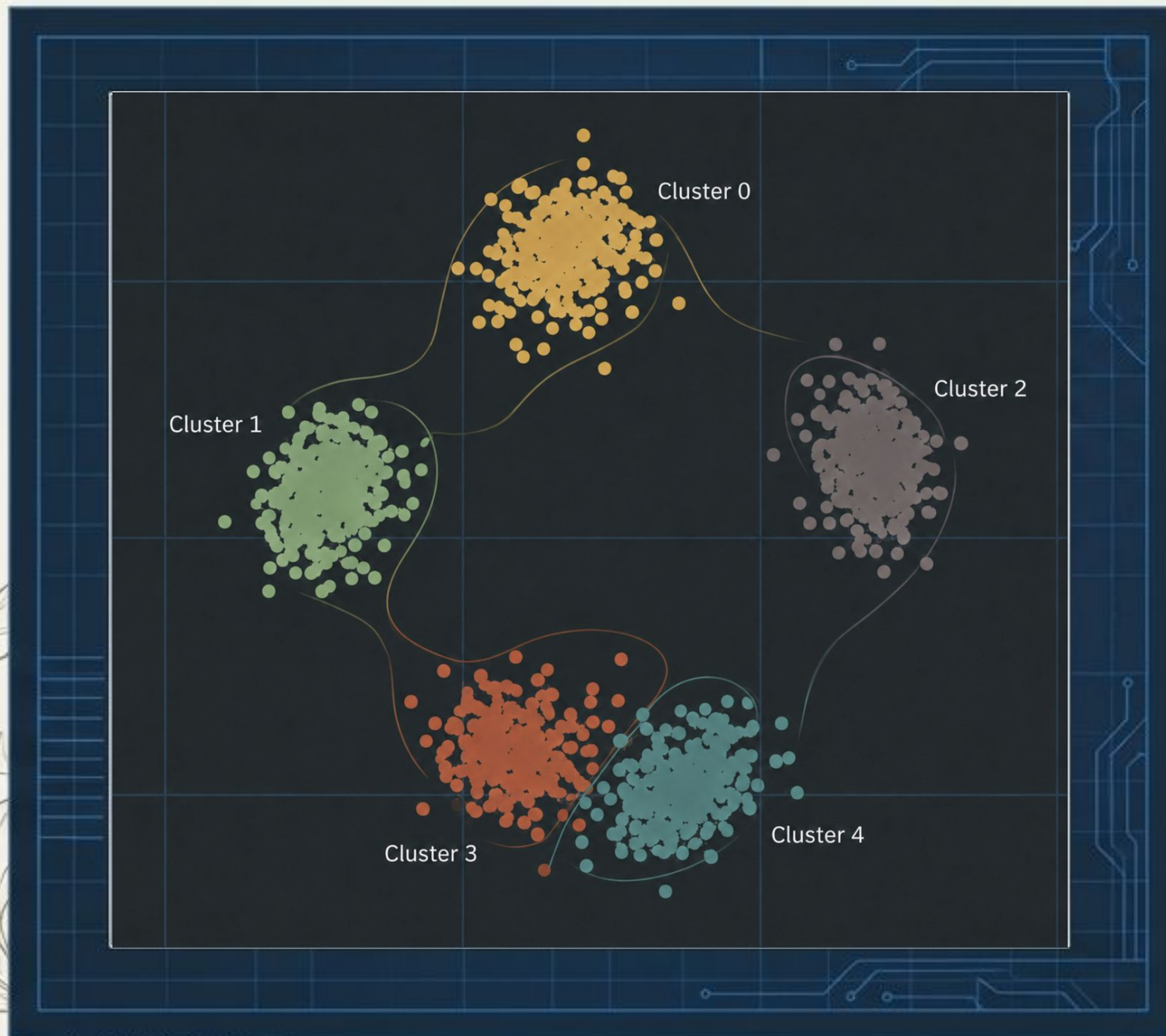
Data Transformation

Raw Measurements

Accipiter albogularis | Mass: 248.8g | Wing Length: 235.2mm

Unsupervised Discovery: Extracting Flight Styles without Labels

Because real-world “flight style” labels didn’t exist in the AVONET dataset, K-Means clustering was deployed to group birds inherently based on feature similarity.

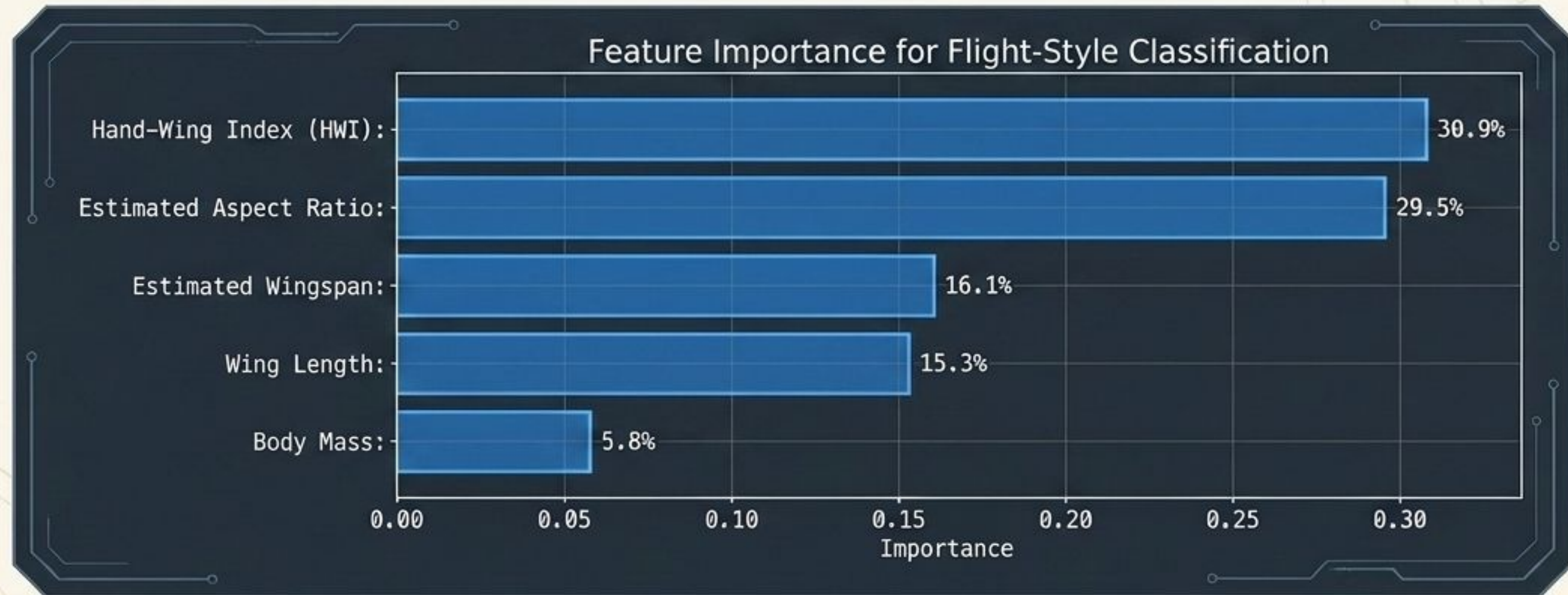


Cluster Mapping Table

Cluster	Biological Interpretation
Cluster 0 & 3 (Soaring / Gliding):	Maps strongly to Accipitridae (hawks/eagles) and Strigidae (owls). Broad wings adapted for riding air currents or slow, silent flight.
Cluster 1 & 4 (Agile Maneuvering):	Maps to Picidae (woodpeckers) and Columbidae (pigeons/doves). Adapted for short, controlled paths through trees.
Cluster 2 (Fast Direct Flyers):	Maps to Anatidae (ducks/geese). Driven by strong, continuous wingbeats.

Supervised Classification & Model Interpretability

Once K-Means defined the five flight-style groups, a Random Forest Classifier was trained to predict these groups for any new bird. The pipeline achieves a 99.6% classification accuracy on the test set.



The AI learned that **wing shape** (HWI, Aspect Ratio) is vastly **more determinative** of flight style than **simple size** or **weight**.

Bridging Unsupervised to Supervised Classification

A Random Forest Classifier was selected to predict the newly discovered morphology-based flight clusters. By ensembling multiple decision trees, the model averages predictions, rendering it highly stable and resistant to noise compared to single-tree methods.



Engine Specs

```
n_estimators = 200
Averages decisions across 200 distinct trees

max_depth = None
Trees grow until optimal leaf purity

test_size = 0.25
75/25 Train-Test Split

random_state = 42
Ensures reproducible modeling
```

Feature Inputs Array

```
mass_kg
wing_length_m
wingspan_m
aspect_ratio_est
wing_loading_N_m2
speed_index
hwi_filled
```


Model Performance & Biological Drivers

99.6%
Test Accuracy

Performance Breakdown

Near-perfect prediction across major clusters.

Cluster 1 (Agile Maneuvering) achieved a 1.00 Precision/Recall/F1 score, with 2051 out of 2054 correct predictions.

Feature Importance (What the AI Learned)

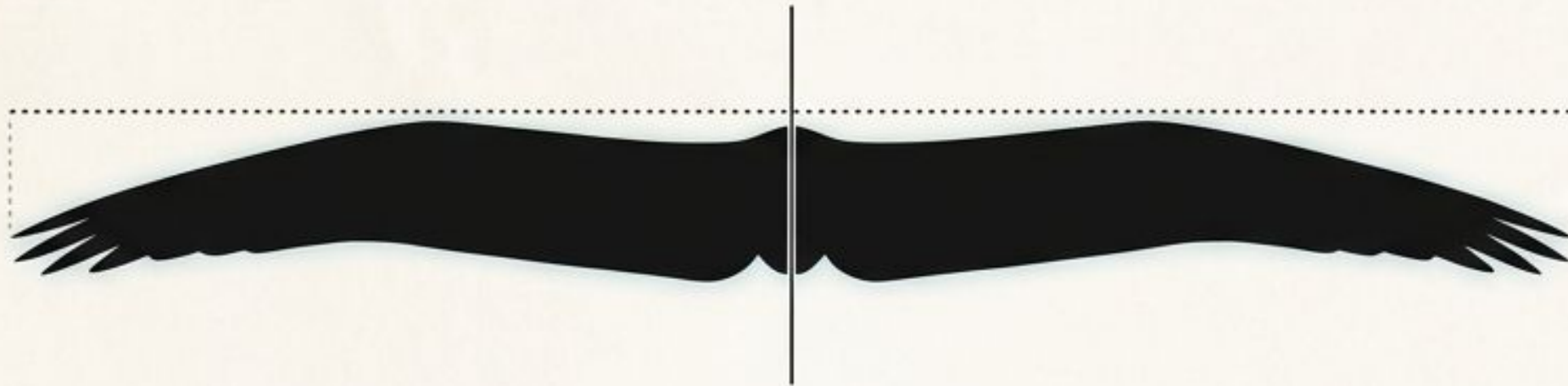


Insight Footer

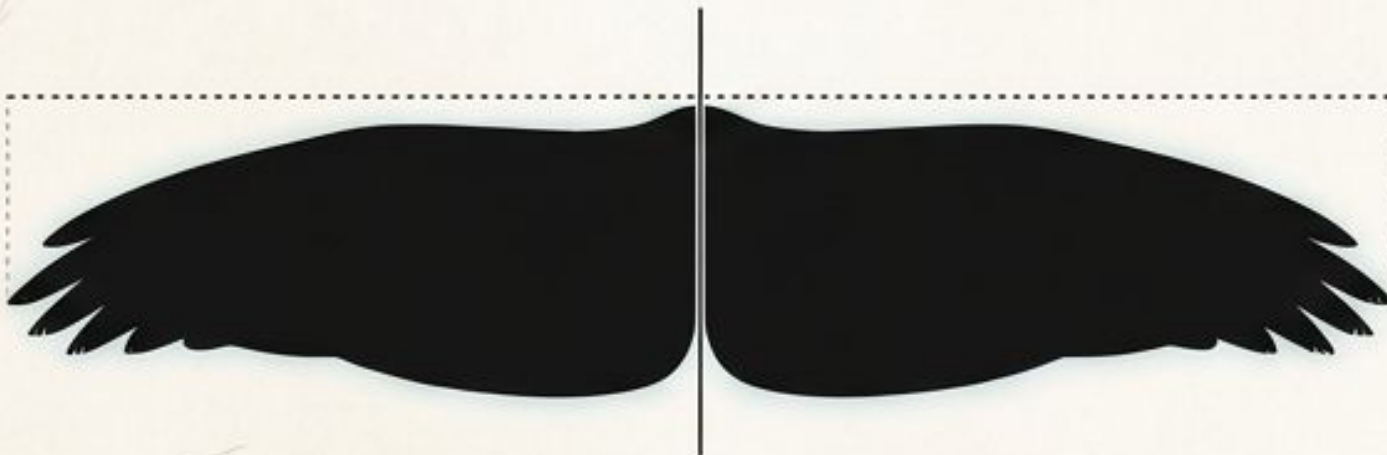
The AI autonomously learned that wing shape (HWI, Aspect Ratio) is vastly more determinative of flight style than simple size or weight.

Generative Visualization: Procedural Wing Modeling

The program features an image generator that translates the AI's continuous morphology values into a top-view, spread-wing planform. Even birds within the same flight cluster produce unique diagrams based on minor feature variations.



AR=15.21 | WL=145.5 | HWI=45.8



AR=8.94 | WL=102.1 | HWI=28.3

The Generative Ruleset:

Larger Estimated Wingspan

→ Longer displayed wings

Higher Aspect Ratio

→ Narrower effective planform

Higher Wing Loading

→ Thicker wing root

Higher HWI / Speed Index

→ Sharper, more swept wingtip

Lower HWI

→ Broader tips, visible feather notches

Explainable AI: Translating Mathematics into English

A true AI pipeline must avoid the 'Black Box' problem. The system evaluates a specific bird's features against global medians to generate human-readable reasoning for its prediction.



The Code

```
def explain_bird(row, global_medians):
    explanations = []

    if row["aspect_ratio_est"] >
        global_medians["aspect_ratio_est"]:
        explanations.append("higher-than-average aspect ratio,
            suggesting longer and narrower wings.")
    else:
        explanations.append("lower-than-average aspect ratio,
            suggesting shorter or wider wings.")

    if row["wing_loading_N_m2"] >
        global_medians["wing_loading_N_m2"]:
        explanations.append("higher-than-average wing loading,
            which often indicates faster or more power-demanding
            flight.")
    else:
        explanations.append("lower-than-average wing loading,
            which can support slower, more efficient flight.")

    # ... (further conditional logic)
```

The Output

Example Case: *Accipiter albogularis*

Predicted Flight Style: Soaring / gliding flyers

System Explanation:

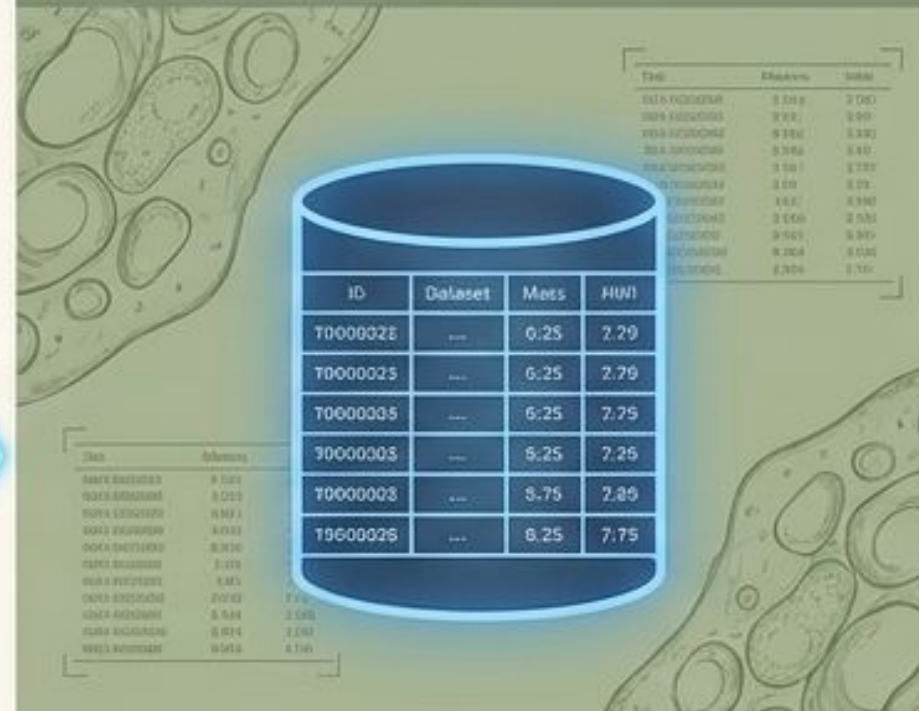
- Higher-than-average aspect ratio, suggesting longer and narrower wings.
- Higher-than-average wing loading, indicating faster or more power-demanding flight.
- Higher morphology-based speed index.
- Relatively high hand-wing index, associated with efficient long-distance flight.

The Multimodal Pipeline in Action: A Complete Example

Image Recognition

```
User uploads an image.  
Computer Vision model  
predicts family: Columbidae.
```

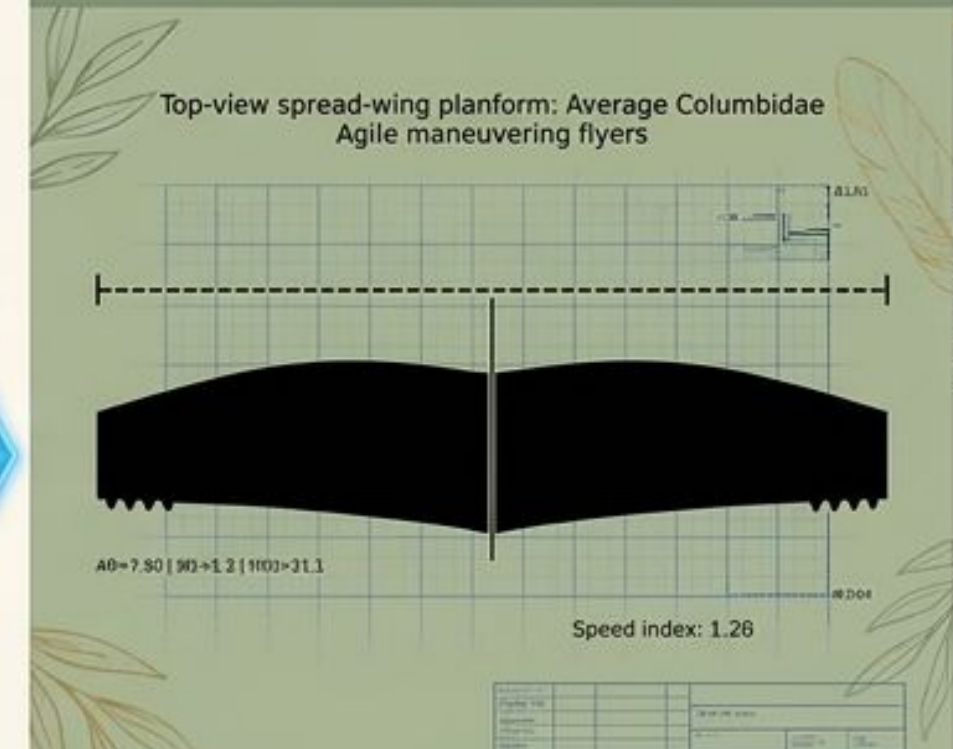
Morphology Integration



System queries AVONET dataset (353 rows used) to build an average Columbidae morphological profile (Mass: 0.25kg, Aspect Ratio: 7.79).

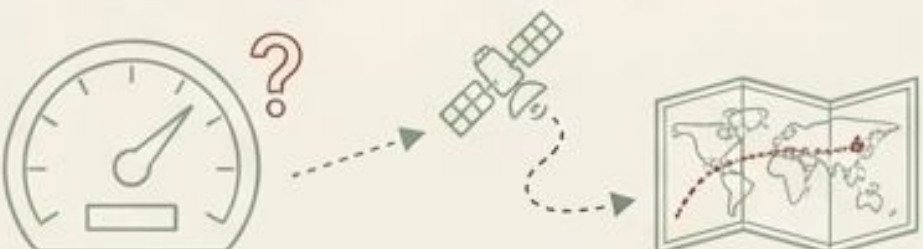
Random Forest predicts flight style: Agile maneuvering flyers.

Generative Output



The system renders the top-view spread-wing planform representing the average structural constraints of the predicted family.

Critical Evaluation: Limitations & AI Ethics

Data Sparsity & Bias	Feature Limitations	Foundational Model Scale
The morphology dataset contained missing values, requiring aggressive filtering. Skewed or sparse training data is the primary driver of biased predictions in ML systems. Image classification quality degrades drastically with blurry or poorly angled photos. 	The morphology-based “speed index” is an engineered mathematical proxy, not a real-world measured flight speed. Future improvements require integrating real GPS tracking or observed migration data to align the model with reality. 	Transfer learning relies on massive foundational models (like MobileNet). As discussed in lecture, these models carry significant environmental compute costs and inherit intrinsic biases from their original, massive, uncurated training datasets. 

Future Explainability: Integrating Grad-CAM

While the tabular morphology data utilizes XAI techniques, the current Convolutional Neural Network (MobileNetV2) remains a black box.



The Grad-CAM Upgrade:

- Gradient-weighted Class Activation Mapping (Grad-CAM) highlights the specific pixel regions that influenced the CNN's prediction.
- Rather than just predicting 'Accipitridae,' the model will visually demonstrate whether it focused on the bird's beak, tail shape, or wing curvature to make its decision, serving as a sophisticated ML debugging tool.

Synthesis: AI as a Biological Translator

Machine learning is not just a tool for optimization; it is a mechanism for discovery. By fusing computer vision with **tabular feature engineering**, this project successfully bridges the gap between identifying what an animal IS and mathematically explaining how it FUNCTIONS.

- Physical structure and behavioral dynamics are mathematically linked.
- Multimodal pipelines provide significantly richer insights than single-model approaches.
- Explainability (XAI) transforms statistical predictions into legible biological intelligence.

Link to code

https://github.com/late8/ai-hw-spring-2026-nn/blob/7b9cd0f8ce109233094add0661d598105c5ac359/MiniRP/bird_flight_multimodal_ai_notebook_classification.ipynb

Youtube demo

https://youtu.be/1pKHqOmS_rA